INCH-POUND

MS16634C
1 May 2012
SUPERSEDING
MS16634B
2 May 1990

DETAIL SPECIFICATION SHEET

RING, RETAINING, EXTERNAL, BOWED "E"
(REDUCED SECTION TYPE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-R-21248.

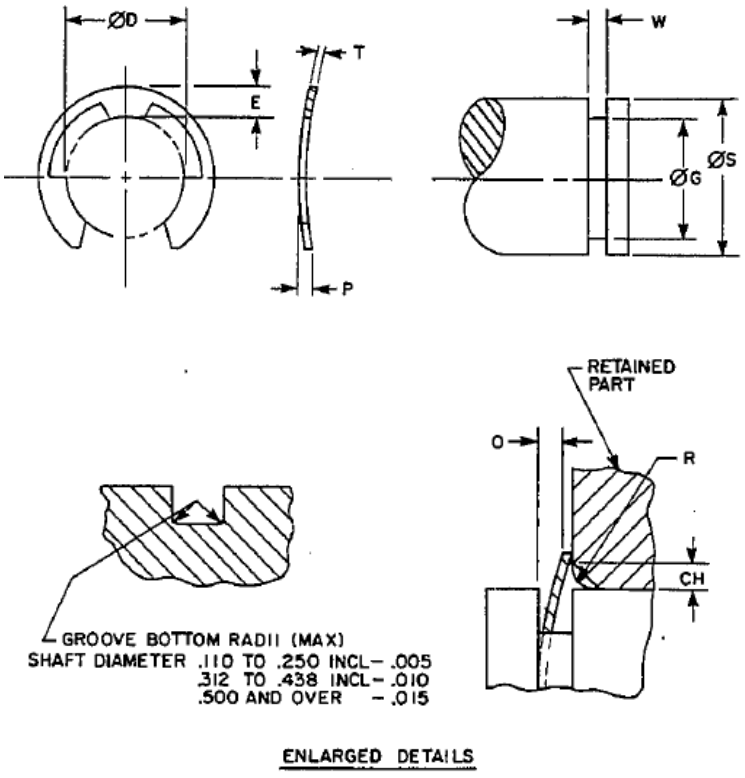


FIGURE 1. RING, RETAINING.

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TABLE 1 DIMENSIONS.

ØS SHAFT (REF)		ØD FREE		E LARGE HEIGHT	T 1/ THICKNESS		P OVERALL BOW HEIGHT		ØG RECOMMENDED GROOVE (REF)		W WIDTH		O 3/ TAKE UP	R 4/ OF RETAINED PART (REF)	CH 4/
INCH	MM	BASIC	TOL	(REF)	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MAX	MAX	MAX
										+.002 -.000 .0015					
.110	2.8	.076	+.002	.155	.010		.030	±.005	.079	FIM 2/	.035		.013	.080	.060
.125	3.2	.094	-.003	.073	.010	±.001	.030		.096		.022		.013	.040	.030
.140	3.6	.102		.090	.015		.033		.105		.025		.014	.060	.045
.156	4.0	.114		.090	.015		.035		.116		.027		.014	.060	.045
.172	4.3	.125		.099	.015		.037	±.005	.127	+.003 -.000	.029		.016	.060	.045
.188	4.8	.145		.101	.015		.038		.147	.003	.030		.027	.060	.045
.219	5.6	.185		.132	.015		.051		.188	FIM 2/	.040		.026	.060	.045
.250	6.4	.207		.167	.025		.058	+.007	.210		.049		.023	.060	.045
.312	7.9	.243		.136	.025		.058	-.008	.250		.047		.023	.060	.045
.375	9.5	.300		.188	.035		.068		.303		.060	+.005 -.000	.022	.065	.050
			+.003			±.002		±.008							
.438	11.1	.337	-.004	.183	.035		.068		.343	+.004	.060		.022	.065	.050
.500	12.7	.392		.213	.042		.084	±.009	.396	-.000	.073		.030	.080	.060
.625	15.9	.480		.240	.042		.089		.485	.004	.077		.035	.080	.060
.744	18.9	.616		.202	.050		.100		.625	FIM 2/	.085		.037	.060	.045
.750	19.0	.547	+.003 -.005	.283	,050		.100	±.010	.580		.085		.037	.085	.065
.875	22.2	.688		.326	.050		.100		.675		.085		.037	.085	.065
.984	25.0	.822		.349	.050		.100		.835		.085		.037	.085	.065
1.188	30.2	1.066		.297	.062		.124		1.079	+.005	.107		.045	.090	.070
1.375	34.9	1.213	+.006 -.010	.348	.062	±.003	.124	±.015	1.230	-.000 .005 FIM 2/	.107		.045	.090	.070

1/ T - THICKNESS "T" APPLIES TO UNPLATED RINGS. FOR CORROSION RESISTANT STEEL AND PLATED RINGS, +.002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i.e. ±.002 SHOULD BE +.004/-.002.

2/ FIM - (FULL INDICATOR MOVEMENT) IS THE MAXIMUM ALLOWABLE DEVIATION OF CONCENTRICITY BETWEEN THE GROOVE AND THE SHAFT.

3/ O - END PLAY TAKE-UP (DESIGN REFERENCE DIMENSIONS).

4/ R AND CH - RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RINGS. THRUST LOADS OF RINGS, RETAINING PARTS WITH CORNER RADII OR CHAMFERS, SEE NOTE (h) ON PAGE 7.

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REQUIREMENTS:

1. CLASSIFICATION: RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE I, CLASS 3 OF THE PROCUREMENT SPECIFICATION.

2. MATERIAL:

(a) CARBON SPRING STEEL, GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A 568/A568M OR ASTM A684/A684M.
(b) CORROSION RESISTANT STEEL, IN ACCORDANCE WITH SAE AMS 5520 (UNS S15700).
(c) BERYLLIUM COPPER, ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH ASTM B194.

3. HARDNESS:

TABLE II. HARDNESS.

ØSHAFT (REF)	CARBON STEEL	CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.110 TO .219 INCL	84.5-87 HRI5N	82.5-86 HRI5N	77-82 HRI5N
.250 TO .312 INCL	66.5-71 HR30N	63-69.5 HR30N	77-82 HRI5N
.375 TO 1.375 INCL	47-52 HRC	44-51 HRC	

4. PROTECTIVE FINISH OR TREATMENT:

a. CARBON STEEL - SHALL BE AS SPECIFIED (SEE TABLE Ill OR IV)

1. CADMIUM PLATE IN ACCORDANCE WITH SAE AMS-QQ-P-416, TYPE II, CLASS 3 OR ASTM B 696, TYPE II, CLASS 5.
2. ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn 5 or ASTM B 695, TYPE II, CLASS 5.
3. PHOSPHATE COAT IN ACCORDANCE WITH MIL-DTL-16232, TYPE Z, CLASS 2.

b. CORROSION RESISTANT STEEL - SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH SAE AMS2700.

5. PART NUMBER: THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE Ill OR IV.

EXAMPLE: MS16634-1011 IS THE PART NUMBER FOR A CARBON STEEL, CADMIUM PLATE, EXTERNAL, BOWED “E” RETAINING RING FOR USE ON A .110 SHAFT DIAMETER.

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TABLE III. DASH NUMBERS FOR MS16634.

Ø S SHAFT (REF)	1/ CARBON STEEL CADMIUM PLATE DASH NO	1/ CARBON STEEL ZINC COAT DASH NO	1/ CARBON STEEL PHOSPHATE COAT DASH NO	STEEL CORROSION RESISTANT DASH NO	1/ BERYLLIUM COPPER DASH NO
.110	-1011	-2011	-3011	-4011	-5011
.125	-1012	-2012	-3012	-4012	-5012
.140	-1014	-2014	-3014	-4014	-5014
.156	-1015	-2015	-3015	-4015	-5015
.172	-1017	-2017	-3017	-4017	-5017
.188	-1018	-2018	-3018	-4018	-5018
.219	-1021	-2021	-3021	-4021	-5021
.250	-1025	-2025	-3025	-4025	-5025
.312	-1031	-2031	-3031	-4031	-5031
.375	-1037	-2037	-3037	-4037	-5037
.437	-1043	-2043	-3043	-4043	-5043
.500	-1050	-2050	-3050	-4050	-5050
.625	-1062	-2062	-3062	-4062	-5062
.744	-1074	-2074	-3074	-4074	-5074
.750	-1075	-2075	-3075	-4075	-5075
.875	-1087	-2087	-3087	-4087	-5087
.984	-1098	-2098	-3098	-4098	-5098
1.188	-1118	-2118	-3118	-4118	-5118
1.375	-1137	-2137	-3137	-4137	-5137

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

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TABLE IV. SUBSTITUTION TABLE, CROSS REFERENCE PART NUMBERS

Ø S SHAFT (REF)	INACTIVE	SUBSTITUTE	SUBSTITUTE	SUBSTITUTE
	CARBON STEEL	1/ CARBON STEEL CADMIUM PLATE	1/ CARBON STEEL ZINC COAT	1/ CARBON STEEL PHOSPHATE COAT
	MS16634	MS16634	MS16634	MS16634
.110	-11	-1011	-2011	-3011
.125	-12	-1012	-2012	-3012
.140	-14	-1014	-2014	-3014
.156	-15	-1015	-2015	-3015
.172	-17	-1017	-2017	-3017
.188	-18	-1018	-2018	-3018
.219	-21	-1021	-2021	-3021
.250	-25	-1025	-2025	-3025
.312	-31	-1031	-2031	-3031
.375	-37	-1037	-2037	-3037
.437	-43	-1043	-2043	-3043
.500	-50	-1050	-2050	-3050
.625	-62	-1062	-2062	-3062
.744	-74	-1074	-2074	-3074
.750	-75	-1075	-2075	-3075
.875	-87	-1087	-2087	-3087
.984	-98	-1098	-2098	-3098
1.188	-118	-1118	-2118	-3118
1.375	-137	-1137	-2137	-3137

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

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NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. RECOMMENDED DESIGN LIMITATIONS AND USAGE.

(a) INTENDED USE- TO PROVIDE UNUSUALLY LARGE SHOULDERS FOR POSITIONING AND MAINTAINING MACHINE COMPONENTS ON SHAFTS. THEY ARE APPLIED RADIALLY AND WITHSTAND CONSIDERABLE THRUST LOAD. THE RINGS ARE BENT LIKE A BOW OUT OF PLANE. FREE ENDS AND OPPOSITE EDGE ABUT THE MACHINE PART, MID SECTION OF RING ABUTS OUTER GROOVE WALL. RINGS WILL TAKE UP END PLAY RESILIENTLY. THEY ARE OF ADVANTAGE WHERE AXIAL ASSEMBLY OF A BORED RETAINING RING IS NOT POSSIBLE. THE USE OF THE FOLLOWING FORMULAS ARE BASED ON THE FACT THAT THE RING WILL NOT FAIL IN COMPRESSION.

LIMITATION ON USE- THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON, ETC.

WARNING - RINGS SHOULD NOT BE OVER EXPANDED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE. IF RING HAS PLAY BETWEEN THE GROOVE DIAMETER AND THE INSIDE RING DIAMETER THIS INDICATES THAT THE RING HAS BEEN OVER EXPANDED, (PROVIDING GROOVE HAS BEEN MACHINED TO RECOMMENDED DIMENSIONS).

FOR APPROXIMATE SAFETY RPM LIMITS SEE TABLE V.

TABLE V. APPROXIMATE SAFETY RPM LIMITS.

Ø SHAFT(INCHES)		.250	.500	1.000
CARBON STEEL AND CORROSION RESISTANT STEEL	RPM LIMIT	25,000	14,000	6,500
BERYLLIUM COPPER	RPM LIMIT	16,000	9,000	---

(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RINGS. ABUTTING COMPONENTS TO HAVE SHARP CORNERS=

$$P = \frac{\pi STX}{3F}$$

WHERE:
P = ALLOWABLE THRUST LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
T = RING THICKNESS (INCHES)
X = ULTIMATE SHEAR STRENGTH OF THE RING MATERIAL (PSI) 1/
F = FACTOR OF SAFETY

A SAFETY FACTOR, F= 3, IS RECOMMENDED SINCE THE RING UNDER LOAD IS SUBJECTED NOT ONLY TO PURE SHEAR STRESSES, BUT ALSO TO BENDING STRESSES.

(c) ALLOWABLE LOAD CAPACITY OF GROOVE WALL

$$P = \frac{\pi GdY}{3F}$$

WHERE:
P = ALLOWABLE COMPRESSION LOAD (POUNDS)
G = GROOVE DIAMETER (INCHES)
d = GROOVE DEPTH (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
F = FACTOR OF SAFETY

TO INSURE A SAFE WORKING LOAD, A SAFETY FACTOR, F= 2, IS RECOMMENDED.

(d) MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT =

$$Z = 2d$$

WHERE:
Z = MINIMUM DISTANCE BETWEEN OUTER GROOVE AND END OF SHAFT (INCHES)
d = GROOVE DEPTH {INCHES)

1/ X = 100,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS UP TO AND INCLUDING .219 INCH SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 120,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .250 AND .312 INCH SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .375 INCH AND OVER SHAFT DIAMETERS OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 110,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF ALL SHAFT DIAMETERS OF BERYLLIUM COPPER.

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(e) LOCATION OF GROOVE =

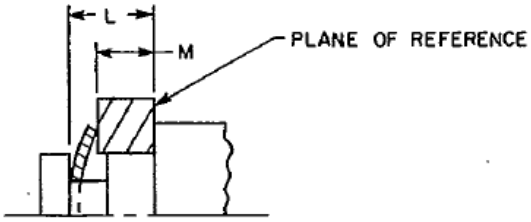
$L_{MAX} = {}^{M}_{MIN} + {}^{T}_{MIM} +$ TAKE-UP (AS LISTED)

$L_{MIN} = {}^{M}_{MAX} + {}^{T}_{MAX} + .005$ 1/

TAKE-UP $\geq {}^{\Delta}L + {}^{\Delta}M + {}^{\Delta}T + .005$ WHERE: ${}^{\Delta}L = {}^{L}_{MAX} - {}^{L}_{MIN}$

${}^{\Delta}M = {}^{M}_{MAX} - {}^{M}_{MIN}$

${}^{\Delta}T = {}^{T}_{MAX} - {}^{T}_{MIN}$



(f) VIBRATION =

A RETAINING RING WILL BE SECURE AGAINST VIBRATION IF ITS SHEAR FATIGUE STRENGTH EQUALS OR EXCEEDS THE FORCE CREATED BY THE MASS OF THE PART BEING REPEATEDLY ACCELERATED AGAINST THE RING.

(1) SHEAR STRENGTH OF RING (CRITICAL). ABUTTING COMPONENTS TO HAVE SHARP CORNERS.

$.36XTS \geq \frac{W}{g} a$

OR FOR HARMONIC MOTION

$a = \delta\omega^2$

$.36XTS \geq \frac{W}{g} \delta\omega^2$

WHERE:

X = ULTIMATE SHEAR STRENGTH OF RING MATERIAL (PSI) 2/
T = RING THICKNESS (INCHES)
S = SHAFT DIAMETER (INCHES)
W = WEIGHT OF ABUTTING PARTS (POUNDS)
g = ACCELERATION DUE TO GRAVITY (IN/SEC^2)
δ = AMPLITUDE OF VIBRATION (INCHES)
ω = ANGULAR SPEED (RAD/SEC)
a = ACCELERATION (IN/SEC^2)

(2) COMPRESSION STRENGTH OF GROOVE MATERIAL (CRITICAL) =

$.55YdG \geq \frac{W}{g} \delta\omega^2$

WHERE:

Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
d = GROOVE DEPTH (INCHES)
G = GROOVE DIAMTER (INCHES)
OTHER SYMBOLS AS SHOWN IN (1)

1/ IT IS ADVISABLE FOR SAFER RING ASSEMBLY TO CALCULATE ${}^{L}_{MIN}$ WITH AN ADDITIONAL .005 IN ORDER THAT THE RING IS NOT FLATTENED OUT COMPLETELY IN ADVERSE TOLERANCE CASES.

2/ X = 100,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS UP TO AND INCLUDING .219 INCH SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 120,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .250 AND .312 INCH SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .375 INCH AND OVER SHAFT DIAMETERS OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 110,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF ALL SHAFT DIAMETERS OF BERYLLIUM COPPER.

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(g) IMPACT CAPACITY OF RING OR GROOVE WALL =

$I_R = \frac{PT}{2}$ FOR THE RING (INCH POUNDS), ABUTTING COMPONENTS TO HAVE SHARP CORNERS.

$I_G = \frac{Pd}{4}$ FOR THE GROOVE (INCH POUNDS).

WHERE:

P = ALLOWABLE THRUST LOAD OF RING OR GROOVE (POUNDS).
T = RING THICKNESS (INCHES).
I_G = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS).
d = GROOVE DEPTH (INCHES).
I_R = IMPACT CAPACITY OF RING (INCH POUNDS).

(h) LOAD CAPACITY, WITH THE RETAINED PART RADIUSED OR CHAMFERED. THE ALLOWABLE THRUST LOADS, WHEN COMPUTED FROM THE FORMULAS ON PAGE 6, WILL NOT BE AFFECTED WHEN (E) RINGS ARE USED AGAINST PARTS HAVING CORNER RADII OR CHAMFERS UP TO THE LISTED MAXIMUM DIMENSIONS.

4. CHANGES FROM PREVIOUS ISSUE. MARGINAL NOTATIONS ARE NOT USED IN THIS REVISION TO IDENTIFY CHANGES WITH RESPECT TO THE PREVIOUS ISSUE DUE TO THE EXTENT OF THE CHANGES.

MILITARY INTEREST

Custodians:
Army - AR
Navy - OS
Air Force - 99

Preparing activity:
DLA - IS

(Project 5325-2012-008)

Review activities:
Army – AV, MI

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